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PALEOMAGNETIC EVIDENCE FOR SPATIALLY DISTRIBUTED POST-MIOCENE ROTATION OF WESTERN WASHINGTON AND OREGON

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Abstract. Anomalous paleomagnetic directions have been determined for 17 sites in the Frenchmans Springs member of the Wanapum basalt formation, Columbia River basalt group. These sites are located in the Ginkgo flows from near Vantage, Washington, to Portland, Oregon, a distance of approximately 300 km. The average paleomagnetic direction for six of these sites, centered around Vantage is D = 147°, I = 41°, α_{95} = 4.5°. The expected Miocene field direction is D = 355°, I = 65°. At some localities there are two distinct Ginkgo flows, in direct stratigraphic succession, with statistically identical anomalous directions. Their anomalous paleomagnetic direction makes these flows a valuable marker horizon in the Columbia River basalt group. The nondipole field direction of the Ginkgo flows correlates well with available results from the Miocene Cape Foulweather basalts of Oregon. This correlation strongly supports the hypothesis that these coastal basalts of Oregon are the distal ends of Columbia Plateau derived basalt flows. The spatial distribution of these anomalous field directions suggests about 14° of clockwise rotation between Vantage and Portland. Combining these data with data from the Oregon Coast basalts allows a maximum declination difference of about 35°. The increase in declination can be best explained by clockwise rotation, about nearby vertical axes, increasing to the southwest across the Columbia Plateau and Oregon coast.

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INTRODUCTION

Recent reviews of the paleomagnetic record from rocks of the leading edge of the North American plate show that many rocks possess paleomagnetic directions which are consistently and systematically discordant [Beck, 1976, 1980]. For western Washington and Oregon this discordance is ordinarily interpreted as evidence that the paleomagnetically studied terranes have rotated clockwise, about a nearby vertical axis, during Tertiary time. Much of the evidence supports the idea that large blocks, such as the Oregon and Washington Coast Range and the Oregon and Washington Cascades, have rotated as near-rigid plates [Simpson and Cox, 1977; Beck and Plumley, 1980; Magill et al., 1982]. Alternatively, Sheriff and Bentley [1980] propose that much of the post-Miocene rotation is due to distributed regional shear. Resolving the relative importance of these two components and putting limits on the eastern extent, chronology and style of this deformation is central

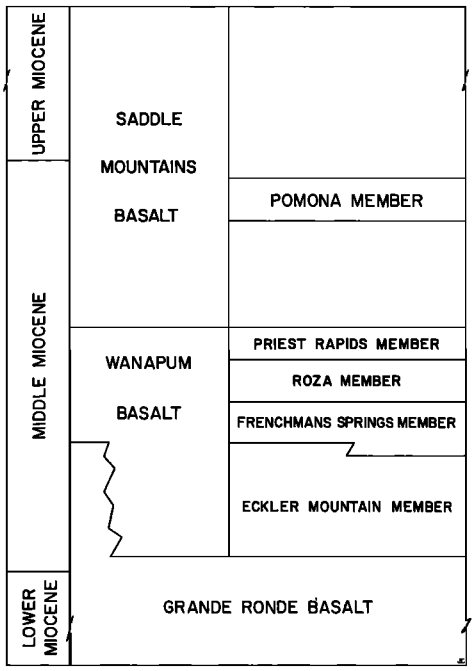


Fig. 1. Stratigraphy for part of the Columbia River Basalt Group. From Swanson et al. [1979a]. Detailed age and polarity date are available in Swanson et al. [1979].

to understanding the evolution of the western segment of the cordillera. In particular, the extent of post-Oligocene rotational deformation has been problematical [Beck and Plumley, 1980; Simpson et al., 1980; Sheriff and Bentley, 1980; Magill et al., 1982].

The basalts of the Columbia River Plateau offer a unique opportunity to measure post-Miocene deformation in parts of Washington and Oregon. The Columbia River basalt group of Washington, Oregon, and Idaho consists of about 2 x 10^5 km^3 of tholeiitic lavas covering an area greater than 2 x 10^5 km^2 [Waters, 1962]. These Miocene flows erupted, from extensive dike systems in eastern Washington and Oregon, over the time period 16.5 mybp to 6.0 m.y.B.P. [McKee et al., 1977; Swanson et al., 1979a]. Swanson et al. [1979a] provide an excellent summary of the basalt stratigraphy and a bibliography of previous work in the area. Beeson and Moran [1979] discuss the stratigraphy of the Columbia River basalts exposed in western Oregon.

The stratigraphic interval of the Columbia River basalt group that is the focus of this investigation consists of the lower flows of the Frenchmans Springs member [Mackin, 1961] of the Wanapum basalt formation (Figure 1). The Frenchmans Springs member is the most extensive member of the Wanapum basalt formation; it usually has three to six flows but can have as many as nine. Geologic criteria including field relations, petrography, and chemistry show that the Frenchmans Springs member exists over a large portion of the Columbia River Plateau (Figure 2). The basalt flows of this member cover a minimum linear distance of 400 km and possibly extend to the Oregon Coast [e.g. Snavely et al., 1973; Beeson et al., 1979]. This paper reports on paleomagnetic results for the Ginkgo flows, the lowermost flows in most of the Frenchmans Springs member. They were extruded at about 14 m.y.B.P. [Swanson et al., 1979a] and thus are of early Middle Miocene age.

I initiated this study of the Ginkgo flows to collect paleomagnetic data from one set of lava flows that extend over a large part of the areal distribution of the Columbia River basalt group. Data of this character should allow a precise determination of relative rotation (if any) between different parts of the

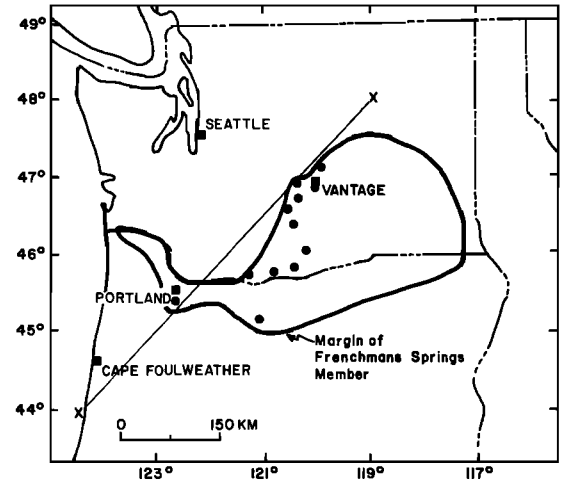


Fig. 2. Distribution of sampling localities for this study with approximate limits of the Frenchman Springs member from Swanson et al. [1979a]. The Frenchman Springs member actually extends to Cape Foulweather. The NE-SW line is the line the paleomagnetic directions are projected onto for Figure 5. See text for discussion.

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TABLE 1. Paleomagnetic Data from the Ginkgo Flows

Site	λ	ϕ	Tilt	N	Dec	Inc	α_{95}	δ	k	H_{opt}
Crescent Bar	47.26	120.05	145-3	5	135.1	48.3	4.9	5.1	200.2	20
Holt Butte	46.96	120.11	75-5	9	147.3	42.9	4.4	6.6	136.5	20
Squaw Creek	46.81	120.36	210-1.5	8	147.0	49.9	4.5	6.1	154.6	40
Getty's Cove	46.84	119.94	140-5	8	145.7	39.0	4.7	6.5	138.0	20
Rest Haven	46.61	120.48	170-10	8	141.7	39.6	6.4	8.7	76.4	20
Union Gap										
Upper	46.53	120.43	180-13	7	148.3	35.3	6.3	7.8	93.5	20
Lower	46.53	120.43	185-12	8	148.7	38.9	7.6	10.3	53.9	40
Rock Creek										
Upper	45.85	120.55	155-4	8	153.9	43.7	8.7	11.8	41.7	20
Lower	45.85	120.55	155-4	8	151.3	51.2	6.4	8.6	77.1	10
Cottonwood										
Creek	45.70	120.59	0	8	155.4	42.7	5.9	8.1	88.5	20
Mary Hill	45.69	120.86	0	7	158.1	42.1	8.7	10.7	49.5	20
Ortley										
Anticline										
Upper	45.64	121.22	140-9	3	147.5	54.0	12.0	6.4	105.9	10
Lower	45.64	121.22	140-9	2	146.1	46.1		4.8		10
Maupin										
*Upper	45.19	121.07	245-7	7	186.1	58.1	16.4	19.8	14.6	60
Lower	45.19	121.07	245-7	3	168.3	51.2	10.3	5.5	145.0	10
Oregon City										
Upper	45.37	122.64	180-3	7	161.0	36.6	13.4	16.4	21.1	40
Lower	45.37	122.64	180-3	8	160.2	42.9	6.3	8.5	79.3	20

λ, ϕ, site latitude, longitude; tilt bearing and plunge of dip vector; N, number of samples; Dec, Inc, site mean declination and inclination; α_{95}, k, Fisher statistical parameters; δ, angular standard deviation; H_{opt}, optimum demagnetization field in millitesla.
*Site not included in further analysis.

Columbia River basalt group over a precisely defined period of time. The Ginkgo flows of the Frenchmans Springs member were an obvious choice for two reasons: (1) these flows outcrop over most of the Columbia River Plateau, in the Columbia River gorge, and in western Oregon, and (2) R. D. Bentley, who is very familiar with their stratigraphy and field identification throughout the outcrop area [e.g., Bentley, 1977; Swanson et al., 1979b] kindly agreed to help with the field work for this project.

Magill et al. [1982] used a similar approach in their paleomagnetic investigation of a slightly younger flow (12 m.y.B.P.) of the Columbia River basalt group. Their results are from the Pomona Member of the Saddle Mountains basalt formation (Figure 1), and its correlative unit, the Pack Sack Lookout basalt, on the Washington coast. Data from this flow suggest that the region of the Pack Sack Lookout basalt has undergone 16° of clockwise rotation since 12 m.y.B.P. [Magill et al., 1982]. The boundary between their rotated western area and nonrotated eastern region is proposed to be within the Cascade Range. Results from the Ginkgo flows (presented here), and some independent geologic evidence [Bentley and Anderson, 1979], suggest to the contrary that the rotation increases southwestward throughout the outcrop area of the Frenchmans Springs member.

PALEOMAGNETIC SAMPLING, DATA COLLECTION, AND RESULTS

The paleomagnetic data presented (Table 1) and discussed in this report consist of 17 paleomagnetic sites from 12

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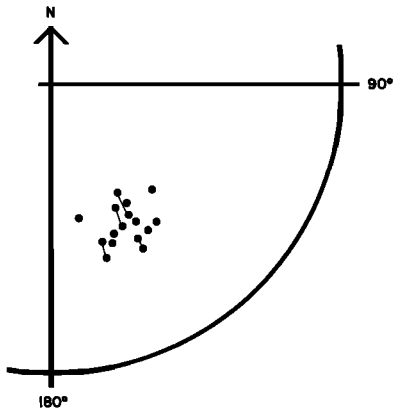


Fig. 3. Lower hemisphere equal-area plot of the 16 acceptable sites of this study. Tie lines connect results from two flows at one locality.

separate Ginkgo flow sampling locations in the Wanapum basalt formation. References for the field criterion for identification of the Ginkgo flows are included in Swanson et al. [1979a]. The geographic locations of the 17 Ginkgo flow sites, spread over a distance greater than 300 km, are shown in Figure 2. With the use of standard paleomagnetic equipment and field techniques [McElhinny, 1973] for sample collection (e.g., rock drills born-of chain saws, in situ sample orientation with magnetic compasses, and back sightings to check for local declination anomalies), three to nine samples were collected at each site, for an average of about seven. Corrections for post-crystallization tectonic tilt are from strike and dip of flow contacts at the sampling localities. The largest tilt correction was 13° but greater than 50% were 5° or less (Table 1). The assumption that the flows had no primary tilt may introduce a maximum error of about 1° which is less than normal error in estimating strike and dip of flow contacts. Thus error in tilt determination is not likely to have contributed significant noise to these data.

Magnetization of the specimens was measured with a Schonstedt SSM-1 spinner magnetometer and specimens were subjected to alternating-field demagnetization tests with a four-axis tumbler system. These stability tests resulted in all specimens being treated to peak demagnetization fields of 5, 10, 20, and 40 mT and some to fields of 60 and 80 mT. There were no consistent secondary overprints. Thus I chose the optimum demagnetization level for each site as the level which maximized the best estimate of Fisher's [1953] precision parameter k. These parameters, along with the paleomagnetic directions for the Ginkgo flows are presented in Table 1. The paleomagnetic directions are plotted in Figure 3.

All but one of the sites of this study responded well to the alternating field cleaning techniques. Values of k, at optimum demagnetization levels, range from 21 to near 300 with an average of about 200 (Table 1). The sample directions from one site contained components of a normal Miocene, or possibly present-day, field direction even at the 80 mT demagnetization level. I excluded this site from further analysis. Eight specimens, from various sites, remained at greater than twice the angular standard deviation from their respective site mean directions throughout the magnetic cleaning process. I also excluded these eight specimens from further consideration. All other specimens reached stable end points (minimal directional change with decreasing magnetic intensity during stepwise magnetic cleaning) and produced site mean directions with acceptably large k (Table 2).

NRM intensities for all specimens were on the order of 10^{-1} emu/cm^3 to 10^{-2} emu/cm^3. The decay of NRM intensity during demagnetization was typical for basaltic rocks containing titanomagnetites as the dominant magnetic phase. Because of the consistent quality of these results, I made a very limited effort, consisting solely of thermomagnetic analyses, to determine the magnetic mineralogy of these rocks.

Figure 4 shows the results from two representative Curie temperature determinations from Ginkgo flow specimens, performed with a thermomagnetic balance set up in vacuums of about 10^{-4} torr and fields of 4-8 kG. For every analysis, I obtained irreversible heating curves and Curie temperatures between 100°C and 200°C. These results confirm that the dominant magnetic mineral in these rocks is an oxidized, high-titanium, titanomagnetite. Unpublished thermomagnetic analyses, with Curie temperatures as low as 50°C, of samples from Columbia Plateau basalt flows other than the Ginkgo flows suggest that the Curie temperatures given above cannot be considered representative

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TABLE 2. Revised Data Used in Figure 5

Site	λ	ϕ	N	Dec	Inc	α_{95}
	Choiniere and Swanson					
1	44.76	124.06	6	167.5	40.8	4.3
2	44.78	124.07	6	172.9	45.1	9.7
	Simpson					
1	44.78	124.07	8	170.9	43.3	4.1
2	44.67	124.07	8	172.9	32.5	6.8

From Choiniere and Swanson [1979] and Simpson [1977]. Revision by R. W. Simpson [personal communication, 1982]; see text for discussion; λ, ϕ are site latitude and longitude; N is number of samples; Dec and Inc are site mean declination and inclination; α_{95} is 95% circle of confidence.

for the whole of the Columbia River basalt group.

Table 1 and Figure 3 show that all the flows presented have paleomagnetic directions that are extremely anomalous when compared to the expected field direction ($D = 356^{\circ}$, $I = 65^{\circ}$) calculated from the Early Miocene paleomagnetic pole for North America [Irving, 1979]. The paleomagnetic directions are considered anomalous because they are about 70° from the expected direction. Thus the Ginkgo flows were most likely magnetized during an excursion or transition of the geomagnetic field during Miocene time. For the sake of clarifying discussion later in this paper, and because geomagnetic field directions vary with position on the earth's surface, I calculated a characteristic mean direction from six sites in a small part of the sampling area near Vantage, Washington. The mean direction of the anomalous field directions from the six sites is as follows: $D = 146^{\circ}$, $I = 41^{\circ}$, $k = 227$. This direction is for comparison with the expected direction given above. The remainder of the anomalous paleomagnetic directions is similar

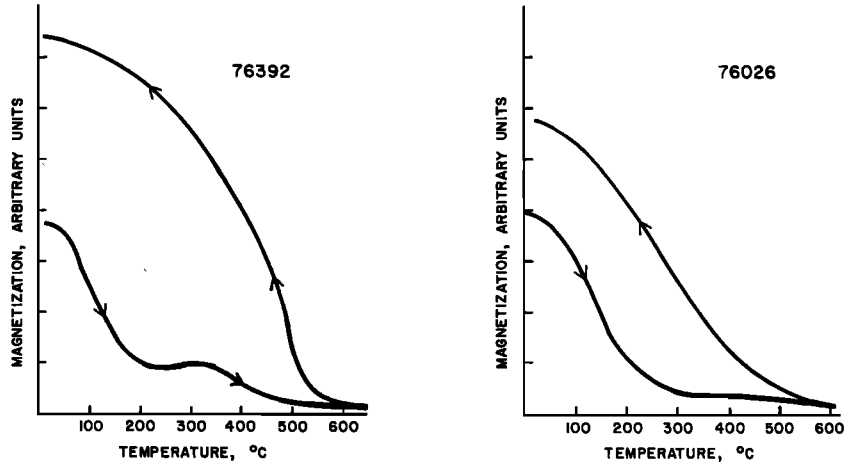


Fig. 4. Saturation magnetization plotted versus temperature for two representative samples from the Ginkgo flows of the Wanapum Basalt. The arrows are relative to the abscissa.

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in their disparity with the expected direction (Table 1). Yet, they also diverge systematically from the mean direction of the six sites in the Vantage area. The nature of this systematic divergence and its structural implications are discussed below.

DISCUSSION

At the time of the sampling program, the anomalous paleomagnetic direction characterizing the Ginkgo flows was unknown. Flows were identified strictly on the basis of geologic evidence. Because of the distinctive nature of the Ginkgo flows' paleomagnetic direction, a more precise identification of Ginkgo flows is now possible. They are now defined as those flows in the lowermost Frenchmans Springs member of the Wanapum basalt formation that have the anomalous paleofield directions of Table 1. At 5 of the 12 localities there are two Ginkgo flows with the characteristic anomalous direction.

Figure 3 is an equal-area stereonet showing the reliable paleomagnetic directions of Table 1; tie lines connect dual flows from one sampling locality. There are only four tie lines on Figure 3 because one of the dual flows gave unreliable results, as discussed in the previous section. Field evidence, including comparison with areas where compound cooling units can be mapped, suggests that the two flows are individual, distinct, lava flows. The top of the lower flow is not weathered nor is there any sediment between the flows. The small angular difference in paleomagnetic directions between the two flows, where both exist at any one sampling locality, strongly suggests a short time lapse between the two flows. Thus there is no evidence indicating a significant time separation between the two flows.

The flows which sandwich the Gingko flows have paleomagnetic directions near expected Miocene directions [Rietman, 1966; Kienle, 1971; S.D. Sheriff, unpublished results, 1980] and the Miocene geomagnetic field had a moderate reversal frequency during Frenchmans Springs member extrusion [Ness et al., 1980]. Therefore it cannot be determined if the anomalous directions result from extrusion during a full reversal of the geomagnetic field or during a period of abnormal geomagnetic field behavior such as a geomagnetic excursion. In either case, the time lag between the two flows is likely much less than 10^3 years [e.g., Fuller et al., 1979]. If the two Ginkgo flows were erupted further apart in time than 10% of a polarity transition, a much greater angular difference between their mean directions would be expected. Evidence for rapid extrusion of large volumes of magma, perhaps of the order of 10^2 km^3 over five days to a week, has also been presented for the Roza Member of the Wanapum basalt formation [Shaw and Swanson, 1970; Swanson et al., 1975]. Voluminous eruptions were characteristic during the evolution of the Columbia River basalt group; therefore the existence of two Ginkgo flows does not necessitate large between-flow time. Thus it is most likely that these two flows flooded the Columbia Plateau in rapid succession. The two Ginkgo flows, with essentially the same anomalous direction, are a valuable and distinctive marker horizon in the widespread Wanapum Basalt Formation. As noted earlier, I sampled these flows from central Washington to near Portland, Oregon, a distance greater than 300 km (Figure 2).

Farther west, on the Oregon and Washington coast, there exist Miocene basalts that are correlative with parts of the Columbia Plateau lavas. Snavely et al. [1973] divided these coastal basalts into three stratigraphic units: (1) the Depoe Bay Basalt, (2) the Cape Foulweather Basalt, and 3) the basalt of Pack Sack Lookout. They show that these units are identical to, as well as correlative and consanguineous with, the Grande Ronde Basalt, Wanapum Basalt, and Pomona member, respectively, of the Columbia River basalt group. Snavely et al. [1973], basing their interpretation on field relations and geologic mapping, concluded that these coastal basalts are of local origin. Alternatively, Beeson et al. [1979], present strong evidence that these coastal basalts are the distal ends of Columbia Plateau-derived lava flows rather than remnants of local volcanism. Beeson et al. [1979] base their interpretation in part on the paleomagnetic directions of the coastal basalts [Simpson, 1977; Choiniere and Swanson, 1979] along with my preliminary results from some of the correlative units of the Columbia River Basalt Group.

Simpson [1977] and Choiniere and Swanson [1979] reported paleomagnetic

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results from the Cape Foulweather Basalt, the coastal correlative of the Wanapum basalt formation. The reported results from these two investigations (Table 2) are very similar to the anomalous field directions from the Ginkgo flows (Table 1).

Choiniere and Swanson's [1979] results from two sites in the Cape Foulweather Basalt are (1) (D, I) = (159.9°, 36.2°) and (2) (D, I) = (198.7°, 53.5°). These directions have been corrected for local dip, the uncorrected directions are (156.2°, 36.7°) and (157.1°, 44.8°), respectively. The smaller angular distance between the uncorrected directions compared to the corrected directions suggests that the local dip represents flow orientation at the time of cooling. Not correcting for local dip seems appropriate considering the reduced scatter and is consistent with the assumption that these lavas are the distal ends of Columbia Plateau-derived basalt flows. The local orientation of the flows are likely a pre-Curie temperature feature resulting from the flows invading and deforming soft, wet sediment [Beeson et al., 1979].

R. W. Simpson [personal communication, 1981] has reconsidered his results [Simpson, 1977] and those of Choiniere and Swanson [1979] for the Cape Foulweather Basalt. These results (Table 2) have not been corrected for local dip. They have been corrected for a regional dip due to the large scale post-Miocene warping of the Oregon Coast Range. Obviously, determining such a tilt correction is no easy nor simple task. These directions (Table 2) are the current best estimates of paleomagnetic directions for the Cape Foulweather Basalt. From my analysis, I agree with Beeson et al. [1979] that the Cape Foulweather Basalt is the distal end of the Ginkgo flows. Complex field relations make absolute delineations of separate cooling units at Cape Foulweather impossible. Thus the four sites in Table 2 are most likely not all independent measures of paleofield direction as recorded by the Ginkgo flows. However, field relations, along with the paleomagnetic directions, suggest both of the anomalous flows are represented.

Figure 5 presents paleomagnetic data from my 16 reliable sites in the Ginkgo flows, four sites from Simpson [1977] and two from Choiniere and Swanson [1979] as discussed above, along with one site each from Rietman [1966] and Kienle [1971]. Data from the other investigations are from localities not sampled during this study. Because my two sites from the Ortley anticline are represented by only five specimens (Table 1), they have been combined and their average declination plotted on Figure 5. Depicted in Figure 5 are the deviations in declination of the 24 sites plotted relative to the Ginkgo flow mean-declination of 147° calculated for the Vantage area. The site locations of relative declination values have been orthogonally projected onto a great circle segment from (48°N, 119°W) to (44°N, 124°W) (Figure 2). After looking at projections onto north-south and east-west segments, I arbitrarily chose this segment to best represent these data. Figure 6 presents these same data on a map of the study area. The most ambiguous paleomagnetic directions are those of Table 2. Because their tilt corrections are poorly constrained, it is very important to note that if these sites are not corrected for regional tilt, as previously discussed, their declinations are close to my farthest west sites at Oregon City, Oregon (Table 1). Increasing the correction for regional tilt increases the amount of apparent rotation between these sites and the others. There is much scatter in these data (Figure 5) but they are all high quality paleomagnetic directions from five independent investigations.

The error bars, which represent values of α_{95} [Fisher, 1953] in Figure 5 are fairly large, and little would be gained by fitting an analytical function to the distribution. Nevertheless, the trend in Figures 5 and 6 is clear and suggests a distinct relationship between site location and paleomagnetic declination. This relationship, with declination regularly increasing to the southwest, can be explained as a fortuitous occurrence, a result of paleofield gradients during cooling of the flows, or as a manifestation of postcooling tectonic effects.

Because the paleomagnetic direction of the Ginkgo flows is from a period of transitional or abnormal geomagnetic field behavior, probably during a time of reduced main-dipole terms, the observed correlation (Figure 5) could be interpreted as a result of paleofield gradients during acquisition of magnetiza-

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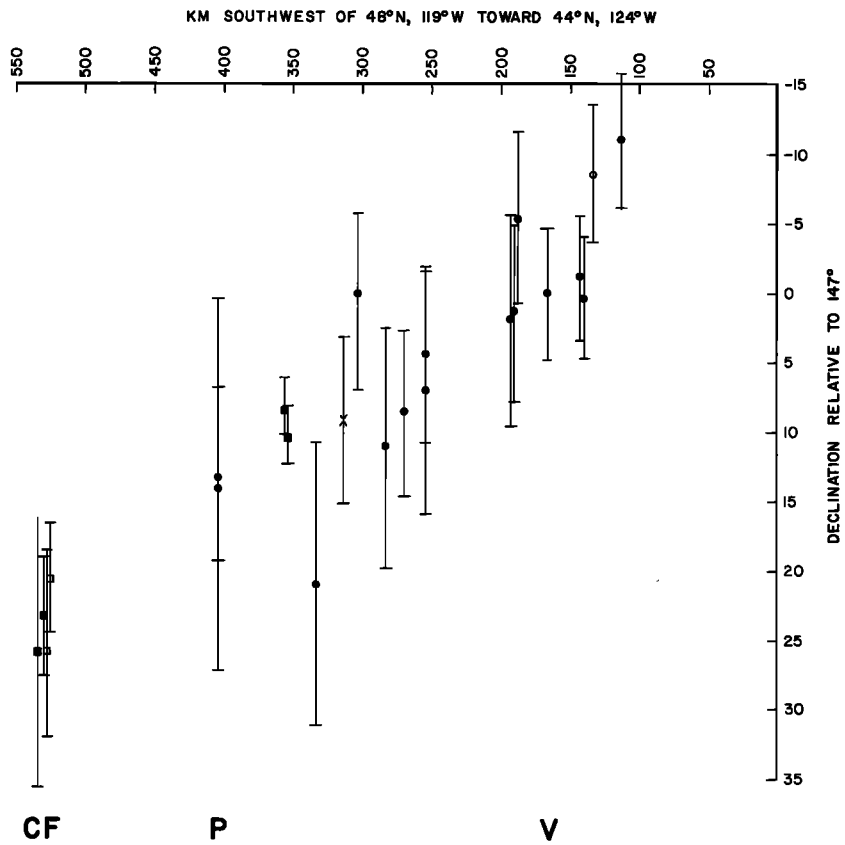


Fig. 5. Declinations of paleomagnetic vectors, plotted relative to 147°; site locations projected onto a great circle from 48°N, 119°W to 44°N, 124°W. Closed circles represent data from my study, open squares from Choiniere and Swanson [1979], closed squares from Simpson [1977], open circle from Rietman [1966], and x from Kienle [1971]. Error bars are α_{95} values of site mean directions. CF, P, and V show the locations of Cape Foulweather, Portland and Vantage, respectively.

tion. Paleofield variation could have produced the Ginkgo flows' magnetization in two different ways. First, the trend apparent in Figure 5 could result wholly from anomalies in the field during magnetization. This requires a reasonable declination gradient of less than 0.1°/km. Second, because the geomagnetic field changes rapidly and erratically during reversals and excursions, the correlation could result from different parts of the flows cooling at different times. Neither of these explanations seem to be the case here because, although the variance in declination and inclination of the sites is similar (Figure 3), the paleomagnetic inclinations do not vary regularly with site location. When the paleomagnetic inclinations are plotted on a north-south, east-west, or northeast-southwest line, or onto a line from the center of the sampling area through a virtual geomagnetic pole calculated on the mean of the anomalous field directions, no systematic variation of the paleomagnetic inclinations is apparent. For a geomagnetic field variation in declination to result in the trend shown in Figure 5, a similar smoothly varying relationship with inclination is most likely. Such is the case for the current nondipole field; every anomaly in declination is associated with a similar anomaly in inclination. This "field gradient problem" exists for any single flow that is sampled over a large area but it is less likely to be significant when the main dipole signal is not attenuated with respect to other field components. It is certainly not a sig-

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nificant component in the data of Magill et al. [1982], and it is very unlikely to be a significant component in the distribution of Figure 5.

An ad hoc or fortuitous explanation, such as systematically accumulated error in tilt corrections, would certainly explain the data (Figure 5), but a tectonic explanation is more scientifically appealing and is consistent with much previous paleomagnetic and geologic work in the western United States [e.g., Beck, 1980; Magill et al., 1982].

As previously discussed, deformation manifesting itself as rotations about near-horizontal axes has been corrected for in Tables 1 and 2 and in Figures 5 and 6. These rotations are easy to recognize and correct for. Rotation about near-vertical axes is not measurable by standard geologic field procedures but can be measured by paleomagnetic techniques. Clockwise rotation, about nearby vertical axes, that increases in magnitude southwestward across the sampling area is the best explanation for the declination-location correlation presented in Figure 5. In the Portland, Oregon, area the Frenchmans Springs member has been rotated clockwise about 14° relative to the mean anomalous direction from the Vantage, Washington, area. Figure 5 suggests that the maximum clockwise rotation around a vertical axis is about 35° when declinations from all sites are compared. The amount of angular displacement increases fairly uniformly across the entire area (Figures 5 and 6). There does not seem to be a distinct grouping of (1) coastal sites versus the others or (2) sites divided by the Cascade Range. This relation is also true if the site locations are projected onto an east-west small circle. These data do not establish an eastern geographical limit on this style of deformation.

Recent interpretations of paleomagnetic data from the Oregon coast [Beck and Plumley, 1980; Magill et al., 1981] treat the Oregon and Washington coast as a coherent tectonic unit. Beck and Plumley [1980] and Beck [1980] interpret their data as showing that rotation of the Oregon coast block ended in the early Miocene. This interpretation is partially based on an early interpretation of some of the data presented in this report [Kienle et al., 1978] that suggested no post-Miocene rotation. Magill et al. [1981] interpret their data as showing that the Oregon coast block continued to rotate as a coherent unit, at a uniform rate, from 20 m.y.B.P. to the present. The results plotted on Figure 5, from five different paleomagnetic studies and from sites well distributed across a large area, suggest that from 14 m.y.B.P. to the present the measured paleomagnetic rotations, in the outcrop area of the Frenchmans Springs Basalt Member, may result from tectonic rotation, around vertical axes, that increases uniformly across the sampling area.

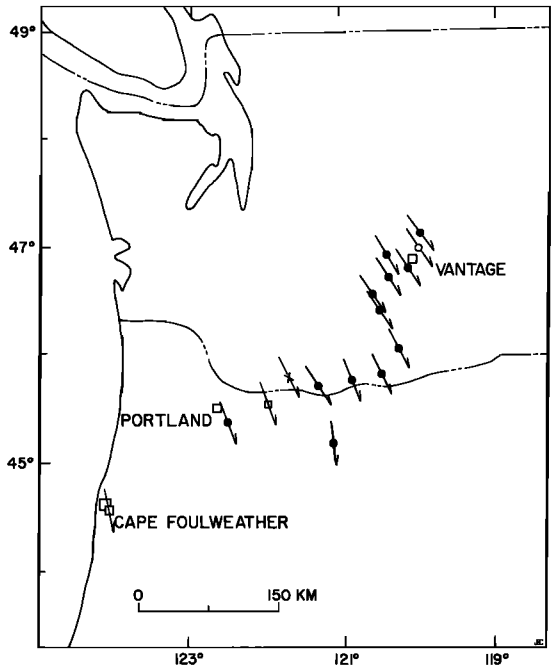


Fig. 6. Declinations of paleomagnetic vectors plotted on a map of their site locations. Each vector has unit length. The average declination is shown for locations with two Ginkgo flows. Closed circles are my data, closed square is average of Simpson [1977] and Choiniere and Swanson [1979], open square from Simpson [1977], open circle from Rietman [1966], and x from Kienle [1971].

My interpretation is in contrast to the single block type of deformation postulated by Magill et al. [1981, 1982], and Simpson et al. [1980] for the same area and time period. The tectonic style suggested by paleomagnetic directions from the Ginkgo flows is also in contrast with the rotation style, of apparently

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coherent tectonic blocks, that occurred previous to 20 m.y.B.P. [Simpson and Cox, 1977; Beck and Plumley, 1980; Magill et al., 1981]. Delineation of the tectonic mechanism prevailing for 14 m.y.B.P. and younger is possible because of the unique situation provided by rocks of the Columbia River Basalt Group: individual lava flows (hence points on a time axis) that extend across the structural terrane in question. Such a rock assemblage does not exist for times before about 14 m.y.B.P. Therefore, it is impossible to paleomagnetically test the hypothesis that the tectonic mechanism I propose for the past 14 m.y. operated before that time and contributed to the observed rotations along the coastal strip.

However, my data are not entirely inconsistent with the idea of a coherent Coast Range block that has rotated since 14 m.y.B.P. The paleomagnetic sites near Portland, Oregon (Figure 2) are not in the proposed Coast Range block [Magill et al., 1982] and the sites near Cape Foulweather, Oregon, are from a small geographic area. That is, the sites from Cape Foulweather represent only one point on the trend shown in Figure 5. Thus, the rotated paleomagnetic directions from those sites do not prove that distributed rotation operated across the Coast Range block. Distributed rotation is suggested by these data because the relative declination appears to increase continuously (Figure 5) across the whole sampling area rather than only stepwise between the Cascade Range and the Coast Range.

Distributed rotation and rotation of a single Coast Range block are end members of a spectrum of plausible tectonic scenarios; many alternative models are possible. The most reasonable geologic model is one with structural domains of size between that of the end members. Such domains are likely of various sizes; some may be as large as the proposed Coast Range block of Magill et al. [1982]. It is quite likely that tectonic jostling of these various domains contributed structural plunge, as well as tilt, to some of the sampled blocks. Thus, although tilt is easy to recognize and correct for, there probably exists some unrecognized plunge at some of the paleomagnetic sites. This problem may contribute a component of the scatter between various points on Figure 5 and leads one to expect imperfect agreement between neighboring points.

There are various ways to produce the rotations shown on Figure 5. In a zone of constant north-directed simple shear, such as might occur with coupling between the Farallon and North American plates, paleomagnetic declinations would be rotated clockwise just as they are in western Washington and Oregon. If the amount of simple shear continuously decreased toward the interior of the two states, this mechanism would explain the observed rotations (Figures 5 and 6). However, no evidence in the rocks suggests the required amount of finite strain. Obviously, the Columbia Plateau basalts as well as the other rock units of Washington and Oregon do not deform as a simple continuum. More likely, crustal blocks of various size have been rotated by a combination of edge interaction as well as by ductile transport on their bases. In this model, the amount of coupling and interaction decreases with distance from the plate boundary. Thus the amount of rotation is also expected to decrease with distance from the plate boundary. This process of distributed shear is similar to that proposed by Beck [1980] to explain tectonic transport of displaced crustal fragments of the Cordillera. However, in this case, the zone of distributed shear is within the Ginkgo flows of the Wanapum Basalt member rather than between discontinuous crustal fragments.

The hypothesis that there is distributed shear throughout the Columbia Plateau, southwestern Washington, and the Oregon coast requires that the structural geology of this region be fairly complex. Structural elements that may have facilitated this deformation are numerous and include (1) abundant north and northwest trending dextral strike slip faults of the Columbia Plateau and adjacent Cascade Range [Bentley and Anderson, 1979] and (2) east to northeast trending folds in the same area [Bentley, 1977, 1980; Campbell and Bentley, 1981]. Movement on these structures has been continuous since 16 m.y.B.P. with major pulses at 14-13 m.y.B.P. and 9-7 m.y.B.P. [Bentley et al., 1980]. The same workers have suggested that a similar structural style existed as early as Late Eocene time. Therefore the tectonic mechanism I propose may have started at that time.

The hypothesis of distributed shear operating on at least part of western North America is not new. Wise [1963]

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proposed a similar idea; a few years later, Atwater [1970] suggested that western North America be considered as a broad, soft boundary between two rigid plates. Beck [1976] proposed that simple shear has contributed to emplacement and/or rotation of many crustal fragments added to the North American Cordillera since the Mesozoic. Beck [1980] and Magill et al. [1982] have contributed and discussed much of the paleomagnetic work performed to test those hypotheses. They conclude that movement of large coherent blocks is the primary tectonic component behind their rotated paleomagnetic directions. The Ginkgo flows, with their anomalous paleomagnetic direction and large areal distribution, provide the necessary paleomagnetic data to support the hypothesis that distributed shear has also contributed to tectonic rotations of the Pacific Northwest.

In summary, paleomagnetic data from the Ginkgo flows of the Columbia River basalt Group suggest the prevailing tectonic mechanism for Columbia River Plateau and Oregon and Washington coast deformation, at least since 14 m.y.B.P., may well be southwestward increasing clockwise rotation about a nearby vertical axis, not a simple rotation of one large tectonic block. The driving force for this deformation is coupling between the North American and Farallon plates.

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